

CT Foundation Models, Attribution and Uncertainty

A controlled study on 3D lung-nodule patches

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Technical project report. This is not a peer-reviewed publication.

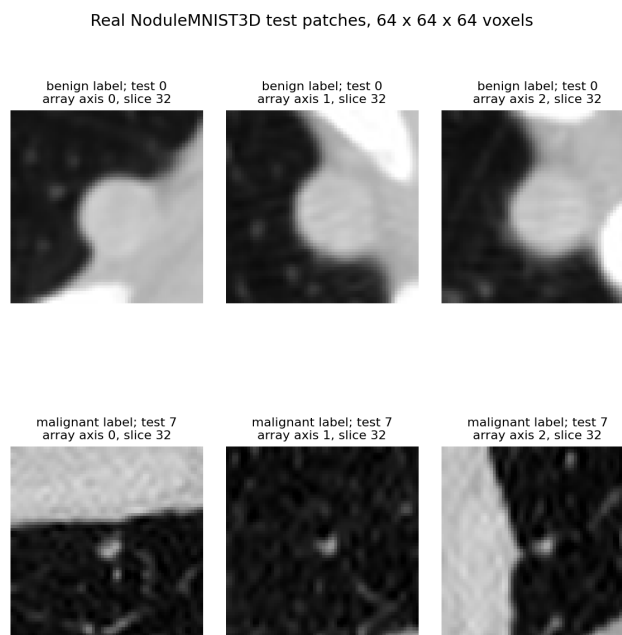
Research question and contribution

Does CT pretraining improve classification when labels are limited, and how do attribution maps behave under perturbation and uncertainty checks? This project fine-tunes the existing CT-FM encoder, compares matched scratch controls, and implements attribution, calibration and uncertainty experiments. CT-FM pretraining was performed by Pai et al.; it is not a contribution of this project.

Summary. Available paired runs show small mean AUC gains from pretraining. The seed-0 full-data AUC is 0.8951 versus 0.8733 from scratch, but a patch-level bootstrap interval for the difference includes zero. Current attribution experiments do not establish faithful or clinically validated explanations.

Real input data and labels

NoduleMNIST3D64 is a MedMNIST v2 benchmark derived from LIDC-IDRI. It groups malignancy ratings 1/2 and 4/5 into two classes and excludes rating 3. The short labels “benign” and “malignant” are **rating-derived, not biopsy-confirmed diagnoses**. Splits contain 1,158 training, 165 validation and 310 test patches. The test set has 246 lower-rating and 64 higher-rating patches.



Rating-derived labels, not biopsy confirmation. MedMNIST v2 / LIDC-IDRI, CC BY 4.0.

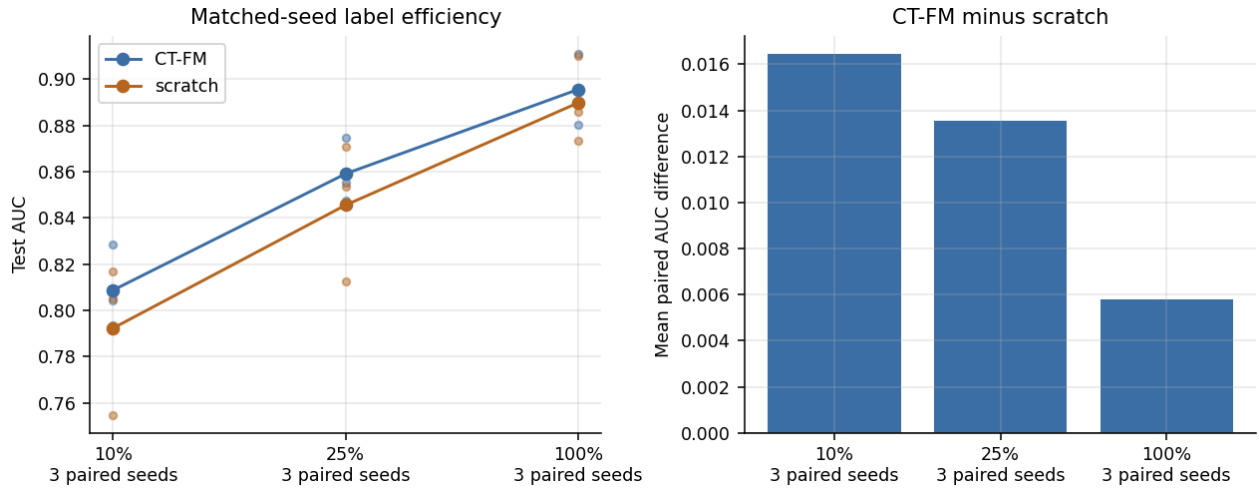
Real test patches 0 and 7, showing slice 32 along each array axis. Anatomical orientation is not established by the NPZ metadata. Source: MedMNIST v2 / LIDC-IDRI, CC BY 4.0.

Model and classification results

The official CT-FM encoder is loaded through `lighter_zoo`. Its deepest 512-channel, 4^3 feature map is pooled and passed through LayerNorm, dropout 0.3 and a linear classifier. The complete classifier has **77,763,042 parameters**. The scratch control reinitialises the same architecture. Inputs are 64^3 uint8 patches scaled to $[-1, 1]$, a benchmark adaptation rather than the original CT-FM full-scan preprocessing.

Matched arms share stratified label subsets for each seed. Training uses class-weighted cross-entropy, AdamW, encoder/head learning rates $10^{-5}/10^{-3}$ and 25 epochs. Validation AUC selects checkpoints. Only seed/budget combinations with both arms contribute to paired averages.

Labels	Patches	Paired seeds	CT-FM AUC	Scratch AUC	Gain
10%	116	0,1,2	0.8087	0.7923	+0.0164
25%	290	0,1,2	0.8591	0.8455	+0.0136
100%	1,158	0,1,2	0.8955	0.8898	+0.0058



Six saved full-data checkpoints were re-evaluated on all 310 test patches:

Model	ROC AUC	Accuracy	Balanced accuracy
Pretrained, seed 0	0.8951	0.8226	0.8304
Scratch, seed 0	0.8733	0.7548	0.7877
Pretrained, seed 1	0.9111	0.8516	0.7678
Pretrained, seed 2	0.8805	0.5839	0.6916
Scratch, seed 2	0.8860	0.8129	0.7839
Pretrained, seed 10	0.9093	0.8774	0.8303

A 2,000-replicate stratified bootstrap of the seed-0 AUC difference gives a 95% percentile interval of $[-0.0184, 0.0659]$. It resamples patches, not patients, and does not include training variability. The seed-2 scratch AUC exceeds the corresponding pretrained AUC. Three completed pairs are recorded at each budget, but the seed-1 scratch checkpoint is unavailable. The seed-1 pretrained record AUC is 0.911077 versus 0.911141 on re-evaluation; the cause of this small discrepancy is unresolved. No universal superiority claim is supported.

Attribution, uncertainty and interpretation

Three Zennit composites and bottleneck Grad-CAM are evaluated on 64 patches. Deletion replaces ranked voxels with the *current batch mean*; insertion restores them from a 5^3 average-blurred input. Stability uses only the first 16 patches, three noise repeats and sigma 0.05.

Method	Deletion AUC	AOPC	AOPC minus random	Stability
Random	0.5870	0.3679	0	n/a
EpsilonPlusFlat	0.8193	0.1466	-0.2213	0.0124
EpsilonGammaBox	0.8224	0.1436	-0.2242	-0.0146
EpsilonAlpha2Beta1	0.8164	0.1494	-0.2185	0.2201
Grad-CAM	0.7121	0.2487	-0.1192	0.9235

Random deletion causes a larger confidence drop than all four attribution methods. Faithfulness is not validated; masking-induced distribution shift remains a hypothesis. The Zennit implementation has no architecture-specific canonizer or validated relevance conservation for the residual and normalisation operations. Finite-map checks on two cases do not resolve that limitation. General conclusions about LRP would therefore be premature. Grad-CAM’s coarse 4^3 origin also helps explain its smoothness; stability is not proof of faithfulness.

Head-only MC dropout uses 20 passes. The 64-case ECE is 0.1453. On all 310 test patches, saved MC-dropout results have AUC 0.8960 and ECE 0.1265; a four-checkpoint probability ensemble has AUC 0.9085 and ECE 0.0505. The ensemble AUC is below the best member’s 0.9111. These are test-set observations, not externally validated calibration claims.

Entropy/AOPC Spearman correlations range from -0.1245 to 0.2214. All four unadjusted two-sided p-values exceed 0.05 (0.079 to 0.327). No clear association is established. Low mutual information from head dropout does not prove aleatoric uncertainty or clinical reliability.

Verification, limits and reproduction

The audit checked data provenance, six full-test checkpoint predictions, ensemble metrics, saved-array statistics and two attribution cases. Paired summaries use completed GitHub records; differing earlier server files are archived. Lower-budget scores and full scratch seed 1 are record-based because those checkpoints were unavailable. Full training and the original stochastic analysis were not repeated.

The distributed arrays lack patient/site identifiers. Split independence and possible overlap with the CT-FM pretraining collection were not independently audited. There is no external cohort, whole-CT evaluation, clinical reader study or validated explanation ground truth. These are the next validation needs, not completed achievements.

Repository, sample volumes, checked weights and reproduction guide

Verification record and raw result files

Sources

MedMNIST v2, Yang et al. (2023): <https://medmnist.com/>

CT-FM, Pai et al.: <https://arxiv.org/abs/2501.09001>

Official checkpoint: [project-lighter/ct_fm_feature_extractor](https://github.com/project-lighter/ct_fm_feature_extractor)

Zennit rule/canonizer documentation: [Rules, composites and canonizers](#)